

What is 'real'? Identifying erosion and deposition in context of spatially-variable uncertainty

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ABSTRACT

The ability to ascribe landscape change to specific processes and events depends on the ability to accurately measure change with confidence. Many studies simply compute the difference between two digital elevation models (DEMs) and do not quantify and propagate vertical uncertainty from multiple sources. Not accounting for the influence of uncertainty on the results represents a particular problem in situations where the uncertainty of one or both surfaces varies across space and the spatial distribution of uncertainty is not consistent through time. Using LiDAR data from 2009 and 2016 for Follets Island, Texas, USA, the purpose of this study is to present a probabilistic approach to DEM change analysis that accounts for spatially-variable uncertainty surfaces and their impact on the computed DEM change. The probability that a computed change is not an artifact of the uncertainty in either surface is determined by computing the area where the two uncertainty distributions overlap. This overlapping area provides information about the likelihood that the observed change is influenced by uncertainty in both surfaces. The described approach is a relatively simple method for propagating uncertainty in one or more datasets; however, it is not an approach to quantifying uncertainty. Rather, this approach relies on uncertainty from one or more sources already having been quantified. Given its adaptability to different datasets, the change probability (CP) approach has potential application in identifying areas of 'real' change in disciplines beyond coastal geomorphology.

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1. Introduction

Changes in elevation and the shape of landforms are commonly used by geomorphologists as a first line of evidence to determine the processes responsible for landscape change. Specifically, evaluating change across space and time provides valuable insight into spatial and temporal patterns of landscape evolution or other phenomena, can be used to identify areas of substantial positive and negative change. The observed change can serve as the foundation for models predicting future conditions, and, thereby, allow managers to determine appropriate hard and soft engineering approaches to protect infrastructure and lives (e.g. Houser et al., 2008, 2015, 2018a; Houser and Barrett, 2009; Houser, 2012; Roberts and Wang, 2012; Eamer and Walker, 2013; Sherman et al., 2013; Heathfield and Walker, 2015; Weymer et al., 2015a). Most of these studies have depended on a simply change analysis, in which the surface change is based on a simple difference between the surfaces.

Digital surface models (DSMs) provide a snapshot of spatial arrangement of a phenomenon at a point in time, in which stacking several DSMs can provide insight into spatiotemporal patterns of change. A specific subset of DSM is the digital elevation model (DEM), used to

explicitly represent elevation data. For example, Houser et al. (2015) used stacked DEMs of elevation above sea level to quantify the rate of foredune recovery following large hurricanes that overwash and completely erode the dune. DSMs can be time-consuming to produce and require hours of manual delineation and validation, but technological improvements in synthetic aperture radar (SAR), satellite-based interferometry, light detection and ranging (LiDAR), and structure-from-motion (SfM) technology enable researchers and managers to collect increasingly greater volumes of surface data, with increasing efficiency and accuracy. Change analysis is traditionally done in a geographic information system (GIS) by subtracting two co-located surfaces at two different times (Mathew et al., 2010; Houser et al., 2013, 2015; Lentz et al., 2013; Sherman et al., 2013; Scott et al., 2016; Burvingt et al., 2017, 2018; Splinter et al., 2018). Specifically, an older surface (t_1) is subtracted from the more recent surface (t_2) for a given area. Areas with negative change values would indicate a decrease in the phenomenon through time while pixels with positive values indicate that the landscape has increased in elevation. Although results of this change analysis may provide valuable insight into patterns of change, the observed change surface ($DSM_{t_2} - DSM_{t_1}$) may be affected by instrument uncertainty (measurement uncertainty) and interpolation uncertainty. Used here, the term 'uncertainty' is preferred over 'error' because it does not require comparison to a higher-accuracy reference value,

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which 'error' directly implies and is not common with many remote sensing datasets.

It can be difficult to determine what is real change and what is within measurement uncertainty and how that uncertainty propagates as more DSMs are compared. A relatively small amount of uncertainty in the elevation of one or more DSMs may result in an overestimation or underestimation of the actual change, as compared to field-based measurements. Overestimating the change or underestimating the uncertainty in either DSM may produce a result with significant change when the actual change is within the measurement uncertainty for one or more surface. As a consequence, uncertainty and uncertainty propagation in DSMs has received increasingly more attention in geomorphology due to its significant potential to affect results and management applications (e.g. Walker and Nickling, 2003; Su and Bork, 2006; Wheaton et al., 2010; Sutherland, 2012; Eamer and Walker, 2013; Walker et al., 2013; Heathfield and Walker, 2015; Anderson, 2019). Drawing conclusions from a simple DSM subtraction without any uncertainty propagation may result in erroneously identifying 'real' changes where the observed change is actually conflated with vertical uncertainty in the two DSMs. Within coastal environments the management implications for misidentifying 'real' change may range from no action taken, where the apparent change appears 'real' and significant when change is actually conflated with uncertainty (i.e. errors of commission, false positives), to beach nourishment and artificial 'dune' construction, where the apparent change appears insignificant and not sustainable when it is actually significant (i.e. errors of omission, false negatives). Whether the errors are errors of omission or errors of commission, misidentifying 'real' change can have long-term implications for the sustainability of a coastal system in context of continued climate change and sea level rise. Vertical uncertainty is challenging to quantitatively address because it may be directly impacted by variations across the landscape, such as slope gradient and vegetation.

Previous research attempted to determine the likelihood that change from one time to the next is 'real' by using a combination of student *t*-test and spatial adjacency through a filtering approach (Wheaton et al., 2010; Heathfield and Walker, 2015) or by applying Moran's *I* test for a pre-determined window size against a global change value (Eamer and Walker, 2013; Walker et al., 2013). However, fine-scale information is overgeneralized and lost when pixels are switched from erosion to deposition or deposition to erosion based on surrounding values within a user-specified area. In areas where a single pixel has been eroded while all surrounding pixels have been aggraded, it is problematic to assume that this change is simply an artifact of the uncertainty at that location. Similarly, fine-scale changes are lost when testing the observed change in a pre-determined window against a global measure of change using Moran's *I*. To assume that a change in one pixel is caused by uncertainty is particularly problematic in highly dynamic coastal and fluvial systems (Walker and Nickling, 2003; Hugenholtz et al., 2009; Heathfield and Walker, 2015; Houser et al., 2015, 2018b; Hapke et al., 2016; Charbonneau et al., 2017; Brenner et al., 2018), where changes may vary in erosion or deposition from one pixel to the next. While the issue of uncertainty propagation has been recognized in fluvial research (Anderson, 2019), it remains poorly addressed in coastal geomorphology. Rip current channels or beach cusps are examples of spatially and temporally dynamic coastal features where some areas may appear to erode or aggrade independently of the surrounding pixels. The spatial variability of uncertainty has not been adequately addressed in highly-dynamic environments.

Not accounting for spatially-variable uncertainty in time-change analyses can have potentially important implications for management and policy decisions in any environment, although these implications can be particularly pronounced in coastal systems. Non-rocky coastal environments, subject to continued sea level rise and human development, are particularly prone to change because they are highly erodible landscapes that can change significantly over multiple space and time scales. Strong wind and waves can cause significant coastal erosion in

hours to days and recovers over several years (Houser et al., 2008, 2015, 2018b; Hapke et al., 2013; Sherman et al., 2013; Hapke et al., 2016; Splinter et al., 2018). Over much broader space and time scales, sea level rise is likely to cause significant coastal flooding and alter alongshore currents, potentially altering sediment transport pathways to, from, and along the coast. Follets Island, Texas is an ideal location to highlight this importance because it is a very 'low island' (i.e. low-relief and elevation), does not have a seawall, and is moderately developed.

The purpose of this paper is to introduce a statistical approach to characterizing change between two DSMs where the uncertainty is spatially variable and dependent on the data source. The change probability (CP) approach introduced here utilizes a normally distributed vertical buffer around each DSM to represent the vertical uncertainty. While the change value is computed by subtracting one surface from the other, the probability of the observed change being real is determined by one minus the overlapping uncertainty distributions integral. Accounting for uncertainty in this way provides a probability that the computed change is not influenced by uncertainty (e.g. the computed change is 'real'). This relatively simple method is adaptable to a wide variety of surfaces and applications, providing a measure of confidence in change analysis in context of uncertainty. The focus on this paper is not in uncertainty quantification, rather it is focused on propagating already quantified uncertainty from one or more sources to compute the probability of 'real' change. This approach is focused on error or uncertainty propagation and only uses thresholding to illustrate how the approach might be used in natural resource management, where identifying specific areas of 'real' change has broader implications. A coastal geomorphology case study is used to demonstrate the utility and importance of propagating vertical uncertainty in change analysis through the CP approach; however, this approach is applicable across a wide range of fields, provided the measurement uncertainty of a phenomenon can be quantified or modelled.

2. Methods

2.1. Uncertainty typology

Uncertainty can be decomposed into three primary sources: (1) data source, (2) data processing, and (3) long-term natural variability (Sutherland, 2012). Understanding how each of these sources contributes to the combined uncertainty of a dataset is essential when seeking to understand patterns of change through space and time. While uncertainty may never be eliminated completely, care should be taken to minimize it in data collection and processing across a range of environments.

Data source uncertainty, such as flight altitude, camera parameters, LiDAR instrument accuracy, or inertial motion unit (IMU) accuracy, is dependent on the platform and sensor used to collect the data. In remote sensing the data source uncertainty for a photogrammetrically-derived DSM is likely different than the source uncertainty for a LiDAR-derived DSM, which is likely different from the source uncertainty for a SfM-derived DSM. The magnitude and structure of data source uncertainty can be systematic (e.g. roll and scanner angle variation due to platform instability, or variation in the output light energy from the LiDAR sensor) or random, depending on the interaction of the sensor and platform with the variability of the surface being measured.

Data processing uncertainty is introduced during data pre-processing and the spatial interpolation processes used to convert the points to a raster surface, or similar structure. For example, filtering LiDAR data by return number or thinning the point cloud to some pre-defined optimal density may remove points that are more accurate while keeping other points that are less accurate. In addition, understanding biases (magnitude and structure) introduced by the interpolation algorithm is vital because an inverse-distance weighting (IDW)

algorithm will produce uncertainty in different locations and magnitudes than an ordinary kriging algorithm with properly fit semivariogram(s). In addition, relatively small differences in the semivariogram sill, nugget, range, and model will affect the uncertainty in the kriged surface. The magnitude and distribution of data processing uncertainty can greatly affect the interpretation of the surface.

The third type of uncertainty in DSMs is due to long-term natural variability. This uncertainty is due to the date and time of data collection in relation to naturally changing environmental conditions, such as tidal fluctuations and turbidity. For example, collecting coastal imagery or LiDAR at high tide can yield a very different surface from that based on data collected at low tide, especially in the swash zone. Even with green-wavelength LiDAR instruments, higher concentrations of suspended solids in the water column will limit the ability of LiDAR to penetrate the water column and introduce vertical uncertainty and, likely cause data gaps.

2.2. Change probability (CP) approach

The approach introduced here employs vertical buffers to quantitatively represent the uncertainty in a DSM (Fig. 1). Each DSM has a corresponding uncertainty raster that contains the values representing the magnitude of uncertainty. While this uncertainty DSM may be represented by a single constant value (Fig. 1a), it may also vary spatially (Fig. 1b), which allows it to more accurately represent real data. The uncertainty rasters, σ_{t_1} and σ_{t_2} , are used to generate probability distributions at each pixel in the DSMs z_{t_1} and z_{t_2} , respectively, where z_{t_1} and σ_{t_1} represent the DSM and uncertainty (standard deviation) at t_1 , and z_{t_2} and σ_{t_2} represent the DSM and uncertainty at t_2 . Using the provided DSMs and corresponding vertical uncertainty distributions, the change in value from t_1 to t_2 at a given location (Δz) can be computed by the following equation:

$$\Delta z = z_{t_2} - z_{t_1} \quad (1)$$

Fig. 1 illustrates these concepts from a profile perspective, with the green line representing the surface at t_1 and the purple line representing the surface at t_2 . Although the uncertainty may be represented as a distribution of constant standard deviation around the surface (Fig. 1a), the proposed approach can also be used with a spatially variable uncertainty value (Fig. 1b). This spatial variability is represented in Fig. 1b by the gradients or varying thickness around each line. It is important to remember, however, that the computed change may be 'real' or affected by uncertainty in z_{t_1} and/or z_{t_2} . Computing the probability of

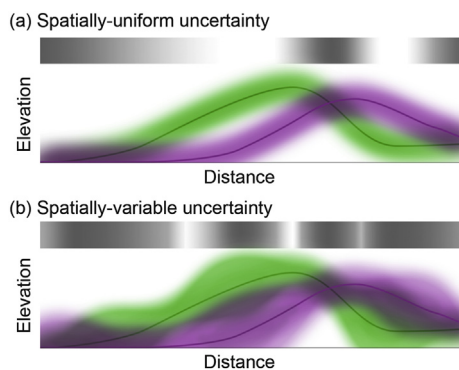


Fig. 1. Profile view of two co-located DEMs where uncertainty above and below each DEM profile (represented as lines) is considered (a) spatially uniform (i.e. the same uncertainty does not vary along each profile line) and (b) spatially-variable (i.e. the magnitude of the uncertainty varies along each profile line). Uncertainty, as represented by the gradients, is modelled as a Gaussian distribution about the DEM value with standard deviation equal to the combined uncertainty for a given location.

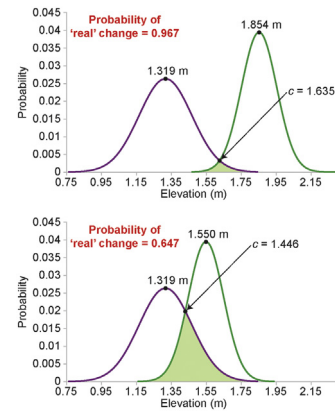


Fig. 2. The change probability (CP) approach is based on the intersection of two error distributions about the DEM value for each pixel. The error or uncertainty distributions have been rotated from vertical distributions to the X-axis for illustration purposes. Probability of 'real' change values range from 0.0 (i.e. observed change is less likely to be 'real') to 1.0 (i.e. observed change is more likely to be 'real'). (a) In cases where the two error distributions overlap very minimally, as indicated by the small shaded area where the two distributions intersect, the probability of 'real' change will be closer to 1.0. (b) In cases where the two error distributions overlap substantially, as indicated by the large shaded area where the two distributions intersect, the probability of 'real' change will be closer to 0.0. Note that although the Y-axis is labelled "probability," it is simply the probability of a value along the X-axis falling within the normal error distribution based on the DEM value (used as the mean) and the computed DEM variance (output from SAGA). In this way "probability" in reference to the Y-axis is not equivalent to the "probability of 'real' change." Probability of 'real' change is computed using Eqs. (2) and (3).

change (P) requires first determining the point where the two uncertainty distributions intersect, c (Fig. 2):

$$c = \frac{z_{t_2}\sigma_{t_1}^2 - \sigma_{t_2}^2 \left(z_{t_1}\sigma_{t_2} + \sigma_{t_1} \sqrt{(z_{t_1} - z_{t_2})^2 + 2(\sigma_{t_1}^2 - \sigma_{t_2}^2) \log\left(\frac{\sigma_{t_1}}{\sigma_{t_2}}\right)} \right)}{\sigma_{t_1}^2 - \sigma_{t_2}^2} \quad (2)$$

where z_{t_1} and σ_{t_1} represent the mean (DSM value) and standard deviation (uncertainty DSM value), respectively, for the normally distributed uncertainty around the surface at t_1 ; and z_{t_2} and σ_{t_2} represent the mean and standard deviation for the uncertainty distribution at t_2 . In other words, z_{t_1} and z_{t_2} can be thought of as the mean values for the uncertainty distributions at t_1 and t_2 . The probability of change is equal to one minus the integral of the overlapping area of the two uncertainty distributions (Fig. 2):

$$P = 1 - [P(x_{t_1} > c) \cap P(x_{t_2} < c)] \\ = 1 - [P(x_{t_1} > c) + P(x_{t_2} < c)] \\ = 1 - \frac{1}{2} \operatorname{erf}\left(\frac{c - z_{t_1}}{\sqrt{2}\sigma_{t_1}}\right) + \frac{1}{2} \operatorname{erf}\left(\frac{c - z_{t_2}}{\sqrt{2}\sigma_{t_2}}\right) \quad (3)$$

where $\operatorname{erf}\left(\frac{c - z}{\sqrt{2}\sigma}\right)$ represents the uncertainty function from the tail to c with the form:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (4)$$

Unlike other approaches that require user-defined thresholds to determine the significance of change at any given location and utilize spatial adjacency to modify the observed/computed change from erosion to accretion or accretion to erosion (Wheaton et al., 2010; Heathfield and Walker, 2015) or apply a Moran's I test over a pre-determined window size against a global change value (Eamer and Walker, 2013; Walker et al., 2013), the calculated change and associated probability of change

from the CP approach are independent of any subjective user thresholds or window sizes. In other words, the CP approach reports the probability of change at all locations instead of only those locations determined to be 'significant.' Furthermore, the CP approach does not use spatial adjacency to modify the observed direction of change based on spatial adjacency, which is likely to overgeneralize fine-scale changes in the landscape.

2.3. Case study: coastal change at Follets Island, Texas, USA

The feasibility and efficacy of the CP approach is demonstrated using a case study from Follets Island, Texas following Hurricane Ike in 2008 and before Hurricane Harvey in 2017. Ideally, both DSMs (September 2009 and September 2016) have a spatially variable uncertainty surface, but this is not always possible depending on the data sources and collection/processing methods. The CP approach is adaptable in that it can accommodate DSMs with spatially-dependent and/or spatially-independent uncertainty surfaces. The case study examines coastal change from 2009 to 2016 using two 1-m resolution LiDAR-derived DEMs, representing elevation, and two 1-m resolution DSMs, where the uncertainty varies spatially due to the spatial interpolation algorithm employed. The reported vertical accuracies of the 2009 and 2016 LiDAR points was 0.15 m and 0.10 m, respectively, although it is important to remember that these values only represent the uncertainty due to instrumentation and do not account for local variation in uncertainty. The 2009 LiDAR point cloud average density was $1.86 \pm 1.03 \text{ m}^{-2}$, and the 2016 LiDAR point cloud average density was $3.16 \pm 2.58 \text{ m}^{-2}$, although it is important to note that the ordinary kriging algorithm used to interpolate the point clouds to DEMs uses information about the statistical structure of elevation values as a function of distance. In other words, the DEMs were produced using fit semivariograms that accounted for spatial structure beyond the individual pixel edges. Both uncertainty surfaces are represented by separate DSMs, although it is possible to have a dual-band raster representing the elevation and uncertainty as separate bands.

Follets Island, Texas, USA is located along the Gulf of Mexico between Freeport, Texas, USA, to the southwest, and Galveston Island, Texas, USA, to the northeast. The island has very minimal topographic relief and a maximum dune height of ~3 m according to the 2016 LiDAR DEM. Because of its minimal relief and elevation, Follets Island is classified as a 'low island' (Duran and Moore, 2013). Several old washover fans are present along the backbarrier side of the island, indicating that the island was breached by previous Hurricanes and other large storms. LiDAR data are freely available for 2009 and 2016 via the NOAA Coastal Topographic LiDAR data portal. Although the 2016 LiDAR survey includes bathymetric data, this paper only utilizes subaerial topographic information that was also included in the 2009 LiDAR survey. Both LiDAR datasets used in this case study were collected by the Joint Airborne Lidar Bathymetry Technical Center of Expertise (JALBTCX).

Follets Island was impacted by Hurricane Ike in 2008 and Hurricane Harvey in 2017; however, it was not affected by any hurricane or tropical storm between the 2009 and 2016 LiDAR surveys. Therefore, any changes (positive or negative) can be attributed to natural post-storm recovery and anthropogenic modification, such as constructed levees or artificial 'dunes.' The amount of recovery of a 'low island', such as Follets Island, can be relatively small (Houser et al., 2015, 2017) and may be difficult to distinguish from the uncertainty associated with each DEM.

2.4. Uncertainty in LiDAR data

While LiDAR uncertainty is often represented as a single value based on the reported instrument accuracy, the uncertainty introduced by the interpolation algorithm may vary spatially, based on the specific algorithm and parameters used for interpolation. For the Follets Island

case study, point clouds were produced using an ordinary kriging algorithm with a fit semivariogram in the System for Automated Geoscientific Analysis (SAGA) software. An advantage of interpolating in SAGA is its ability to produce a variance raster, which varies from pixel to pixel depending on how closely the estimated DEM value aligns with points falling within the pixel area. This variance raster was used as a measure of interpolation uncertainty for demonstration purposes.

Combining the measurement uncertainty with the interpolation uncertainty produces an uncertainty raster that is spatially-dependent. Instrument measurement uncertainty (e.g. uncorrelated random uncertainty or spatially-independent uncertainty) and interpolation uncertainty (e.g. systematic uncertainty or spatially-dependent uncertainty) were combined on a per pixels basis as the square root of the sum of squares, consistent with GIS and remote sensing literature when combining errors (Hodgson and Bresnahan, 2004; Congalton and Green, 2009; Anderson, 2019):

$$\sigma = \sqrt{\sigma_m^2 + \sigma_a^2} \quad (5)$$

where σ is the uncertainty for a given pixel, σ_m is the uncertainty due to instrument measurement uncertainty, and σ_a is the uncertainty due to the ordinary kriging interpolation algorithm. Although the sources of uncertainty used here are uncorrelated, it is possible to expand Eq. (5) to include a correlated uncertainty term or additional uncertainty terms (see Anderson, 2019). It should be noted that the long-term natural variability (i.e. morphological) uncertainty is absent from Eq. (5) because quantifying the impact of slope on uncertainty without introducing user subjectivity in terms of slope thresholds is beyond the scope of this paper.

Accounting for spatially-dependent and spatially-independent uncertainty is essential to appropriately interpreting changes from one time to another. Since both the 2009 and 2016 DSMs were interpolated from LiDAR data, they both have spatially-independent uncertainty, which is useful in demonstrating the ability of the CP approach ability to accommodate spatially complex uncertainty structures. Evaluating change from 2009 to 2016 demonstrates that the CP approach is not limited to surfaces with global accuracy measures or spatially-dependent accuracies but can handle multiple independent uncertainty structures.

The combined uncertainty raster only accounts for two sources of uncertainty; however, additional sources of uncertainty may be quantified and included in the combined uncertainty estimate by adding terms to Eq. (5). Spatial variations in the vegetation community and slope will affect the uncertainty of a LiDAR measurement at any given location. While shrub and forest vegetation tend to overestimate elevation, grassland communities tend to underestimate elevation (Su and Bork, 2006). An additional consideration when quantifying uncertainty due to vegetation is the leaf structure within the community. For example, energy from the LiDAR instrument is more likely to penetrate a community composed of coniferous vegetation because the leaves are needles and have a smaller cross section; therefore, the LiDAR energy is less likely to be interfered with by contacting a leaf. In contrast, leaves on deciduous vegetation have a great cross-section area and are more likely to interfere with the incoming LiDAR energy.

Spatial variations in the LiDAR uncertainty may also be associated with changes in slope across the landscape, where point measurement accuracy generally decreases with increasing slope gradient (Hodgson and Bresnahan, 2004; Su and Bork, 2006). This relationship may be attributed to the vector of LiDAR energy from the instrument to the landscape and the reflection angles depending on the off-nadir sampling angle. Increasing the slope gradient reduces the likelihood that a pulse will be reflected directly back to the instrument and, depending on DSM spatial resolution relative to the slope angle, increases the variability of elevation measurements within a given pixel. Although it is possible to quantify additional sources of uncertainty, the current paper utilizes the spatial interpolation uncertainty and instrument

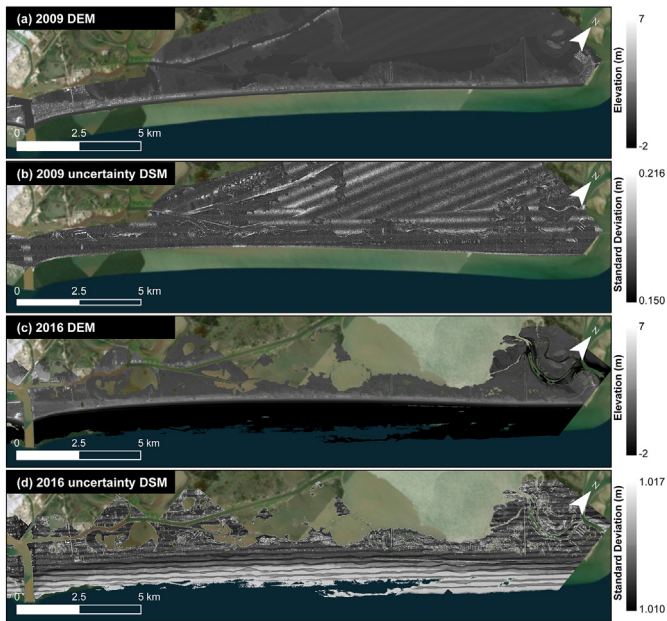


Fig. 3. Follets Island DEMs for (a) 2009 and (b) 2016 with the corresponding uncertainty DSMs (e.g. standard deviation) for (b) 2009 and (d) 2016. Distinct striping in the variance DSMs (b and d) corresponds to overlap between flight lines in the raw LiDAR point clouds.

measurement uncertainty for demonstration purposes because the combination of these sources produces an uncertainty estimate that is spatially-dependent and more realistic than a single global measure of uncertainty (i.e. the instrument measurement uncertainty frequently reported in LiDAR data). In theory, the uncertainty attributed to the spatial interpolation process at least partially includes uncertainty due to spatial changes in slope and vegetation, although neither are explicitly

included in the uncertainty quantification process. Including additional sources of uncertainty would increase the likelihood that the approach can be used to differentiate ‘real’ change from change conflated with uncertainty.

3. Results

The 2009 and 2016 uncertainty DSMs exhibit distinct striping patterns roughly oriented parallel to the shoreline (Fig. 3). This striping extends throughout both datasets and is likely due to variations in point cloud density where flight lines overlap, which are caused by variations in the inertial motion unit (IMU), global navigation satellite system (GNSS) positioning, and the inertial navigation unit (INU) while the platform is moving along a given flight line. Stripes in the 2016 uncertainty DSM are narrower and ‘wavier’ (i.e. cross-shore flight line position varies alongshore) than the 2009 uncertainty DSM, which is a product of the point cloud density stemming from the number of flight lines required to cover the island. Since the 2016 LiDAR point cloud is much denser ($3.2 \pm 2.6 \text{ m}^{-2}$ on average) than the 2009 LiDAR point cloud ($1.8 \pm 1.0 \text{ m}^{-2}$ on average), the 2016 survey required more flight lines to cover the island. In addition to the striping, both uncertainty DEMs are more heterogeneous where flight lines overlap, possibly due to IMU, INU, and GNSS variations from one flight line to the next.

When the 2016 Follets Island DEM is subtracted from the 2009 DEM, the island exhibits variable change (Fig. 4), suggesting that the landscape changed in the seven years elapsed between LiDAR surveys. Red hues represent negative values where land surface elevation has decreased (i.e. erosion), and blue hues represent positive values where the land surface elevation has increased (i.e. deposition). The minimum, maximum, and mean apparent change from 2009 to 2016 is -43.11 m , 8.79 m , and $-0.11 \pm 0.95 \text{ m}$, respectively. Although the mean change is slightly erosional, the relatively high standard deviation suggests caution should be used in interpreting the raw change map.

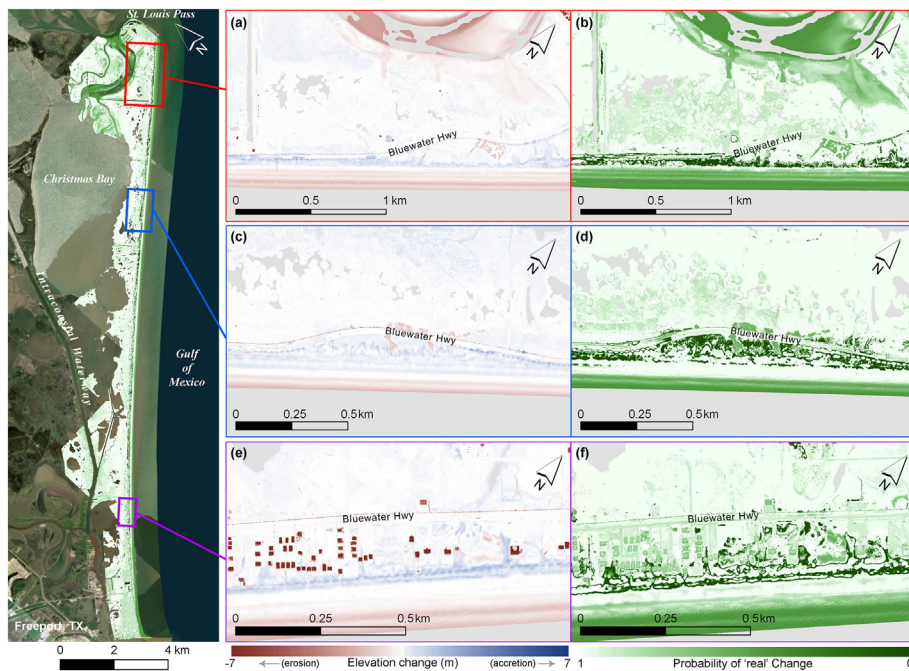


Fig. 4. DEM change from 2009 to 2016 without accounting for uncertainty (a, c, and e) and the probability of change (b, d, and f). Mapping the probability of change, in context of the observed change, provides insight into how uncertainty affects the observed change. (a and b) The northeast end of the island has some human development but has a very dynamic back-barrier tidal inlet system. The probability of change is greater along dynamic back-barrier tidal inlet channels than it is throughout much of the back-barrier flats. (c and d) The central part of the island is relatively sparsely developed and has some of the largest dunes on the island. The greatest probability of change values in the central part of the island, with minimal human development, are concentrated along the dune slopes, while the back-barrier flat has a relatively low probability of change. (e and f) The southwestern part of the island is characterized by houses, powerlines, and other anthropogenic infrastructure. Despite greater human development in the southwestern part of the island, the greatest probability of change values is not along anthropogenic structures but are along the slopes of dunes and depressions.

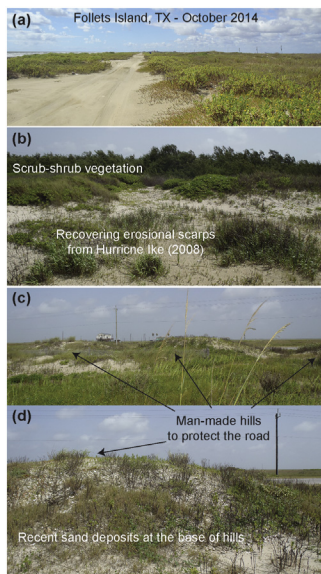


Fig. 5. (a) Photos of Follets Island highlight the very low relief topography of the beach-dune system. (b) Vegetation varies across the island with the tallest dunes in the center of the island supporting a dense scrub-shrub assemblage. Erosional scarps presumably formed by floodwater rushing away from the mainland during Hurricane Ike exhibit evidence of geomorphic (fresh aeolian deposition) and vegetative recovery during 2014 fieldwork. (c and d) Man-made hills were constructed along the southern side of the road to protect the road from future storms. Lighter unvegetated patches of sand visible around the base of these hills are indicative of recent aeolian deposition, although this deposition is limited to the base of the hills and does not extend up to or past the mid-slope.

Potential change probability surface values range from 0.0 to 1.0, with 0.0 representing areas where the two uncertainty distributions overlap substantially, and change is a product of the uncertainty in both surfaces represented by lighter areas in Figs. 4b, d, and f, 5, and 7.

Values approaching 1.0 are areas where the probability of change is increasingly less influenced by the uncertainty in both surfaces represented by darker green areas in Figs. 4b, d, and f, 5, and 7. The mean probability change surface value is 0.11 ± 0.19 .

The greatest 'erosion' appears to occur in built environments where houses and power lines, present in the 2009 data, had been filtered out of the 2016 data during processing by JALBTCX, prior to becoming publicly available. Presumably, the pronounced localized erosion in built areas would exhibit a lower magnitude of change if buildings had not been filtered out of the 2016 data; however, it is not possible to test this hypothesis with the available data. Although these built environments may appear as localized erosional hotspots that are substantial in the raw change DEM, they have low to moderate change probability values (~ 0.1 to ~ 0.4 ; Fig. 5). In contrast, areas adjacent to the man-made structures exhibit less change from 2009 to 2016 but have greater change probability values. When extreme changes in the built environment are filtered out, several heterogeneous erosional hotspots remain in areas with scrub-shrub vegetation. Much of the backbeach (e.g. the area inland of the shoreline where sediment is beyond the reach of daily waves and sediment is likely subject to Aeolian transportation) and foredune system appears to increase in elevation, along with man-made hummocks along the southern edge of the Bluewater Highway, the main road running along Follets Island.

4. Discussion

Results demonstrate that the CP approach is able to compute the probability that change in DSM surfaces is 'real' and not influenced by the spatially-variable uncertainty (Figs. 4b, d, and f, 5, and 7). Much of the island exhibits low to moderate change values, and the most extreme change values occur in built environments where buildings present in the 2009 LiDAR data had been pre-processed and removed from the 2016 LiDAR data (Figs. 4 and 5). Areas with the greatest probability of change are not directly related to the magnitude or direction of elevation change, highlighting the importance of spatially variable

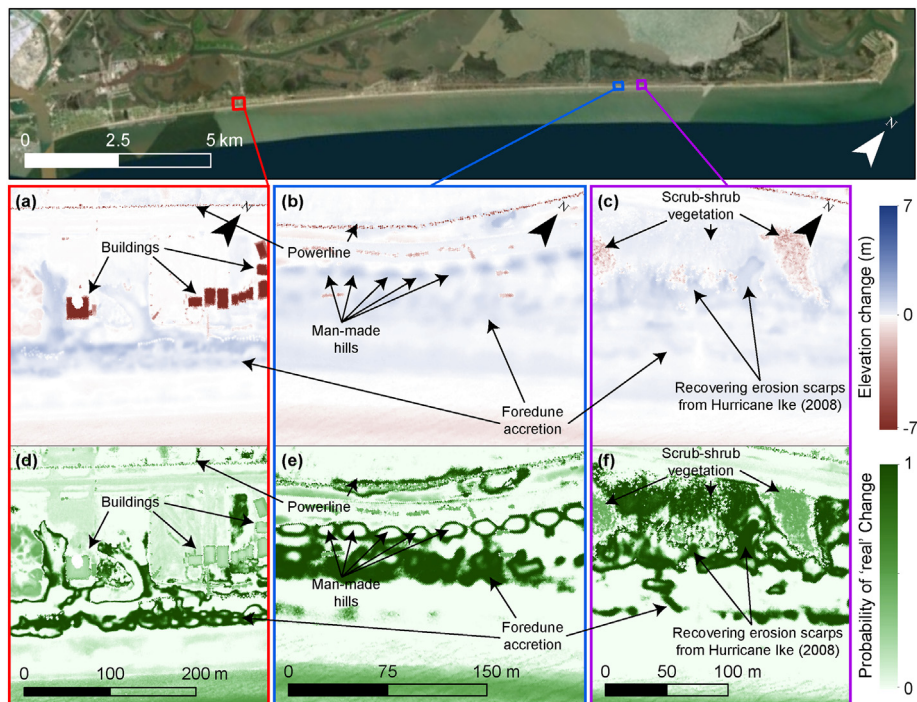


Fig. 6. Important features of note along Follets Island. (a and d) Houses, powerlines, and other man-made structures appear as pronounced areas of erosion in the DEM change surface and have very high probability of change value. (b and e) A series of regularly-spaced hills were built between 2009 and 2016. These hills appear as rings in the probability change surface, where the greatest probability of change is along the edge of the hills. (c and f) Areas of erosion along the foredune slopes have very high probabilities of change in comparison to the backbeach and tops of dunes, suggesting that the probability of change is capturing the process of recovery and sediment accumulation along the dune slope.

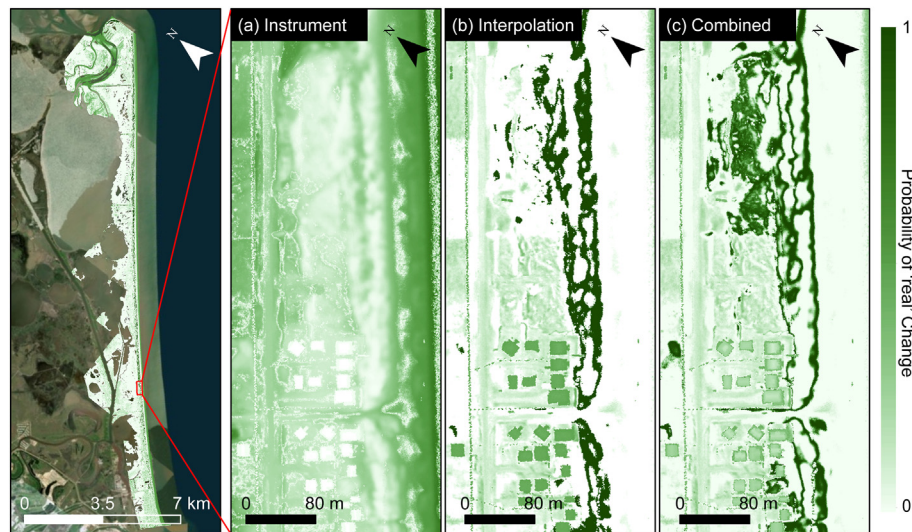


Fig. 7. The probability change surface will be directly impacted by the uncertainty surface(s) included in the analysis. (a) Only accounting for instrumentation uncertainty suggests a large area of the landscape has moderate probabilities of 'real' change. (b) Interpolation uncertainty only highlights buildings and most of the dunes along the island as having high probabilities of 'real' change. (c) Integrating instrumentation and interpolation uncertainty provides more comprehensive information about the probability of 'real' change across the landscape.

uncertainty in DEM differencing and the probability of 'real' change from one time to the next.

When built environments are ignored, the greatest changes in the elevation across Follets Island occurred along the foredune, along man-made hills south of the Bluewater Highway, and throughout the backbeach (Figs. 4 and 5). In 2009, the seaward face of the foredune and the backbeach has a similar morphology to the Bolivar Peninsula after Hurricane Ike in 2008 (Sherman et al., 2013), suggesting that it was eroded due to flood water scouring the beach and dune as it receded back to the Gulf of Mexico. In the 2016 DEM, these erosional scarps appear to be recovering (see the blue hues labelled as "recovering erosion scarps" in Fig. 5), which is further supported by the high probability of change values along the stoss side of the foredune (i.e. the windward side of the dune, on the side where winds approach the dune, often more gently sloping than the 'back' of the dune; the 'stoss slope' faces the Gulf of Mexico in Fig. 4b, d, and f). While using raw elevation change as an indicator of recovery (Fig. 4a, c, and e) suggests that the change is substantial and these erosional scours have recovered, the change probability map (Fig. 4b, d, and f) is more variable, with the greatest probabilities concentrated along the lee slopes of the foredune (i.e. the 'back' of the dune, opposite of the approaching wind direction, often steeper than the windward side of the dune; the 'lee slope' faces away from the Gulf of Mexico in Fig. 4b, d, and f).

Similar to the foredune, changes in beach morphology may be misleading and should be cautiously interpreted. The beach generally increases in elevation from 2009 to 2016 (represented by blue hues in Figs. 4 and 5), and the relatively featureless backbeach morphology in 2009 appears to be replaced by a foredune complex in the 2016 LiDAR DEM (labelled as "foredune accretion" in Fig. 5). These changes suggest that the beach has recovered substantially between 2009 and 2016; however, moderate to low probability of change values across the beach indicate that the observed changes to backbeach morphology are conflated with some uncertainty. Given the very low relief and very gently slopes of the beach in 2009 and 2016, limited dune formation close to the shoreline may be explained by waves planing off any developing dunes from the beach during relatively minor storms, which can reach the base of the foredune. The variable recovery of the backbeach and foredune system, as indicated by the combination of DEM change and probability of change surfaces, highlights the potential impact of not accounting for uncertainty in time-change analyses. Without propagating the uncertainty, it is reasonable to assume that the

system is recovering; however, identifying areas and patterns of 'real' recovery is more complex with uncertainty propagation.

Vertical growth of the back-barrier system is a naturally occurring process and vital to maintaining a viable ecosystem in context of continued sea level rise. It is typically a very slow process occurring over years to decades but may be accelerated through overwash deposits from storms. Much of the Follets Island back-barrier system exhibits very moderate changes from 2009 to 2016 (Fig. 4), as expected in a slow changing and low-relief landscape. Interpreting the observed changes as 'real' and using this change to inform management policy decisions is problematic, as the instrumentation error alone is 0.15 m and 0.10 m for 2009 and 2016, respectively. Accounting for instrumentation and interpolation uncertainty, the probability change surface strongly suggests that the observed change is likely conflated with uncertainty, and further work should be conducted to determine whether the observed changes and patterns are 'real.'

Man-made hills south of the Bluewater Highway (Fig. 6), new foredune accretion in the backbeach, and apparent recovery along the lee slope of the foredune (Fig. 5) align with current understanding of Aeolian sedimentation and coastal recovery/evolution. While the tops of dunes appear to have a variable probability of change, slopes along foredunes and man-made hills have higher change probability values. While it is possible that uncertainty due to slope could affect the probability of change on sloping areas such as the foredune and man-made hills, it is unlikely that the high probability of change values observed along the foredune and man-made hills is the solely due to the lack of slope uncertainty since the observed changes are consistent with coastal Aeolian processes. It is likely that moderately sloping areas, such as the mid- to toe-slope of the dune or man-made hill, exhibit high change probability values because dominant onshore winds transport sediment inland from the recovering backbeach and deposit this sediment where wind velocity decreases, such as around dune-building vegetation or debris along the foredune toe. Given that Follets Island was not affected by a major storm between the 2009 and 2016 surveys, it follows that this continued Aeolian transport caused sediment to accumulate preferentially along the slopes of dunes and man-made hills. In contrast, flatter areas of the beach, tops of hills and dunes, and the back-barrier either receive relatively little sediment or lack vegetation and other debris to reduce wind speed and cause sand deposition. In other words, variations in the change probability surface along the hillslopes accent the ability of the CP approach to identify

and highlight the process of 'real' Aeolian recovery. Fieldwork photos from 2014 support quantitative evidence of recovery along the toe- and mid-slope of the man-made hills and erosional dune scarps (Fig. 6).

In addition to highlighting the importance of uncertainty propagation in time-change analyses, results demonstrate that incomplete characterization and quantification of uncertainty from multiple sources can have a substantial impact on how the DEM change values are interpreted. Fig. 7 highlights the importance of quantifying uncertainty from multiple sources when conducting DEM change analysis. Only accounting for uncertainty due to instrumentation (Fig. 7a) suggests that a large area of the island has a moderate to high probability of experiencing 'real' change. Applying the instrument measurement uncertainty as a single global measure of uncertainty while neglecting the interpolation uncertainty generally underestimates of the amount of uncertainty and overestimates the probability of 'real' change at a given location. In contrast, only accounting for uncertainty due to the interpolation process appears to minimize more moderate change probabilities and emphasize more areas of more extreme change, such as man-made structures and a large proportion of dunes (Fig. 7b). When the instrumentation and interpolation uncertainties are combined (Fig. 7c), the greatest change probabilities occur along the slopes of depressions and hills. Variation in the uncertainty DSMs and propagation of this uncertainty to the change probability DSMs highlight the importance and potential impact of accounting for uncertainty from multiple sources.

While it is common to state the vertical uncertainty due to instrumentation, often as a global measure of uncertainty in the data source metadata, no previous studies include the spatially-variable uncertainty due to interpolation and propagate the combined uncertainty to results. Simply propagating the vertical uncertainty reduces errors of commission and errors of omission because of it accounts for some uncertainty; however, the current paper demonstrates that not accounting for multiple sources of uncertainty can have a substantial effect on spatial patterns in the probability of 'real' change from one time to the next (Fig. 7). Regardless of the spatial interpolation algorithm (e.g. inverse-distance weighting, spline with tension, ordinary kriging, and universal kriging), the distribution of vertical uncertainty will be location dependent based on structure in the data source. Integrating as many sources of uncertainty as possible minimizes the likelihood that errors of commission and errors of omission are present and maximize the likelihood of identifying 'real' change.

Unlike other approaches to change analysis that require user-defined thresholds to determine significance or flip erosion and deposition values based on adjacent pixels (Wheaton et al., 2010; Heathfield and Walker, 2015), the CP approach does not require any subjective user input and does not modify the observed change patterns based on adjacent values. By simply subtracting the two DEMs and reporting a probability of change across the entire area, the CP approach simply provides all the information to the user. How the change values are interpreted in context of the probability of change surface is entirely dependent on the project goals and objectives. The probability of change threshold required to interpret an area as having 'significantly changed' will directly impact the amount of area perceived as 'significant change'. To illustrate how changing the minimum threshold would affect the amount of area identified as 'real' change, a series of theoretical thresholds were set in Fig. 8; however, it is important to note that the authors are not suggesting any minimum change thresholds. More conservative change estimates may utilize a higher probability of change threshold, which will reduce the area interpreted as 'significant change.' In contrast, utilizing a lower probability of change threshold will increase the number of pixels interpreted as 'significant change.' For the 2009–2016 Follets Island change analysis, the amount of area interpreted as significant change does not increase linearly with increasing change probability threshold but increases rapidly at lower change probability thresholds, and very gradually at higher change probability thresholds. Since the decision to apply a minimum probability change threshold is dependent on the project goals and objectives, the current paper does

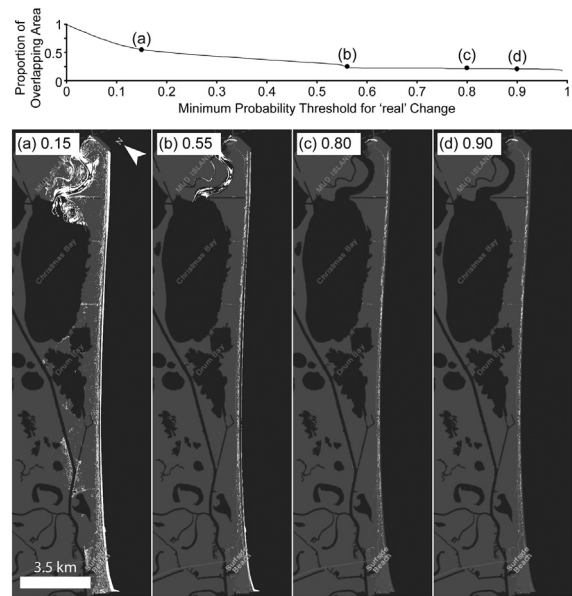


Fig. 8. The proportion of the total area exceeding a minimum probability of change threshold decreases as the threshold value increases. Setting an appropriate probability of change threshold will vary depending on the study objectives and will directly affect the amount of area classified as 'real' change. The white pixels on each map are those exceeding the user-defined threshold indicated for that map and represent areas interpreted as 'real' change. Applying (a) 0.15, (b) 0.55, (c) 0.8, and (d) 0.9 probability change thresholds demonstrates that lower threshold values are more liberal indicators of change and will highlight more area as 'real' change (a and b). In contrast, minimum change probability thresholds approaching 1.0 are more conservative indicators of change and will substantially reduce the amount of area exhibiting 'real' change, as defined by the specific project (c and d).

not recommend any minimum probability of change value but acknowledges that such a threshold would inherently directly affect the results and conclusions.

While the case study used here is based on changes in landscape elevation in the coastal environment, the CP approach to determining the probability of 'real' change is applicable to other mappable phenomena including geophysical measurements (Weymer et al., 2015b; Weymer et al., 2016) and changes in sea ice (Gordon and O'Farrell, 1997; Lindsay and Zhang, 2005; Kwok and Rothrock, 2009; Stammerjohn et al., 2012; Johannessen et al., 2016). Since change analysis is used extensively as a decision support tool and uncertainty can have a potentially important impact on the perceived change, it is important to account for multiple sources of uncertainties. The proposed pixel-based approach is not dependent on any user-defined thresholds and remains true to the data (i.e. it does not switch erosional values to depositional or vice-versa). The portability and efficiency of the CP approach makes it highly useful for researchers, planners, developers, and managers as a way of accounting for uncertainty in their analysis and ultimately improve the accuracy of the final products and resulting management and policy decisions.

5. Conclusion

The CP approach to DSM change detection is a relatively simple approach to propagating uncertainty in change analysis that has several advantages over other existing approaches. First, unlike other methods that rely on user-defined thresholds to determine what changes are 'significant' (Wheaton et al., 2010; Eamer and Walker, 2013; Heathfield and Walker, 2015), the CP approach is independent of any arbitrary thresholds, although a user could apply a minimum change threshold specific to their objectives. The probability change surface provided by the CP approach relies on a normally distributed uncertainty around each DSM to compute the probability that the change at any given location

is 'real' or not. Second, since the probability of change is not dependent on simulations, it is not sensitive to specific simulated values or distributions of values that may skew the probability of change. Finally, the CP approach does not artificially modify very small pockets of relatively minor erosion or deposition to the opposite sign based on adjacent pixels, as do other approaches (Walker et al., 2013; Heathfield and Walker, 2015). While it may be tempting to reverse the sign of erosional pixels surrounded by depositional pixels (or *vice-versa*), to do so assumes that erosion or deposition in one pixel must be spatially adjacent to deposition or erosion, respectively, in an adjacent pixel. In coastal and fluvial geomorphology, in particular, this assumption is flawed, as sediment may be transported more than the distance from one pixel to the next. The CP approach does not make this assumption and retains the integrity of the originally calculated change. It is important to note that the CP approach is not a way to quantify uncertainty, rather it is a way to propagate uncertainty in one or more datasets to determine the probability that the observed change is 'real' and not a product of data uncertainty. Because the CP approach is independent of subjective user-inputs and remains true to the data, this approach has the potential to affect management in highly dynamic environments, such as coastal and fluvial, by providing valuable information about the degree to which the observed change is 'real' or conflated with uncertainty. Given the potential impact of uncertainty on change analyses, care should be taken to quantify as many sources of uncertainty as possible when monitoring or mapping environments or phenomena such as coastal beach and dune systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Anderson, S.W., 2019. Uncertainty in quantitative analyses of topographic change: error propagation and the role of thresholding. *Earth Surf. Process. Landf.* 44 (5), 1015–1033.
- Brenner, O.T., Lentz, E.E., Hapke, C.J., Henderson, R.E., Wilson, K.E., Nelson, T.R., 2018. Characterizing storm response and recovery using the beach change envelope: Fire Island, New York. *Geomorphology* 300, 189–202.
- Burvingt, O., Masselink, G., Russell, P., Scott, T., 2017. Classification of beach response to extreme storms. *Geomorphology* 295, 722–737.
- Burvingt, O., Masselink, G., Scott, T., Davidson, M., Russell, P., 2018. Climate forcing of regionally-coherent extreme storm impact and recovery on embayed beaches. *Mar. Geol.* 401, 112–128.
- Charbonneau, B.R., Wootton, L.S., Wnek, J.P., Langley, J.A., Posner, M.A., Souza, L., 2017. A species effect on storm erosion: invasive sedge stabilized dunes more than native grass during Hurricane Sandy. *J. Appl. Ecol.* 54 (5), 1385–1394.
- Congalton, R.G., Green, K., 2009. Positional accuracy. In: Congalton, R.G., Green, K. (Eds.), *Assessing the Accuracy of Remotely Sensed Data*. CRC Press, pp. 19–52.
- Duran, O., Moore, L.J., 2013. Vegetation controls on the maximum size of coastal dunes. *Proc. Natl. Acad. Sci. U. S. A.* 110 (43), 17217–17222.
- Eamer, J.B.R., Walker, I.J., 2013. Quantifying spatial and temporal trends in beach–dune volumetric changes using spatial statistics. *Geomorphology* 191, 94–108.
- Gordon, H.B., O'Farrell, S.P., 1997. Transient climate change in the CSIRP coupled model with dynamic sea ice. *Mon. Weather Rev.* 125, 875–907.
- Hapke, C., Brenner, O., Hehre, R., Reynolds, B.J., 2013. Coastal Change From Hurricane Sandy and the 201–2013 Winter Storm Season: Fire Island, New York.
- Hapke, C.J., Plant, N.G., Henderson, R.E., Schwab, W.C., Nelson, T.R., 2016. Decoupling processes and scales of shoreline morphodynamics. *Mar. Geol.* 381, 42–53.
- Heathfield, D.K., Walker, I.J., 2015. Evolution of a foredune and backshore river complex on a high-energy, drift-aligned beach. *Geomorphology* 248, 440–451.
- Hodgson, M.E., Bresnahan, P., 2004. Accuracy of airborne LiDAR-derived elevation: empirical assessment and error budget. *Photogramm. Eng. Remote. Sens.* 70 (3), 331–339.
- Houser, C., 2012. Feedback between ridge and swale bathymetry and barrier island storm response and transgression. *Geomorphology* 173–174, 1–16.
- Houser, C., Barrett, G., 2009. Bed elevation changes in the upper-swash zone. *J. Coast. Res. (SI(56 (Proceedings of the 10th International Coastal Symposium)))*, 64–68.
- Houser, C., Hapke, C., Hamilton, S., 2008. Controls on coastal dune morphology, shoreline erosion and barrier island response to extreme storms. *Geomorphology* 100 (3–4), 223–240.
- Houser, C., Labude, B., Haider, L., Weymer, B., 2013. Impacts of driving on the beach: case studies from Assateague Island and Padre Island National Seashores. *Ocean Coast. Manag.* 71, 33–45.
- Houser, C., Wernette, P., Rentschlar, E., Jones, H., Hammond, B., Trimble, S., 2015. Post-storm beach and dune recovery: implications for barrier island resilience. *Geomorphology* 234, 54–63.
- Houser, C., Bishop, M., Wernette, P., 2017. Short communication: multi-scale topographic anisotropy patterns on a Barrier Island. *Geomorphology* 297, 153–158.
- Houser, C., Barrineau, P., Hammond, B., Saari, B., Rentschlar, E., Trimble, S., Wernette, P.A., Weymer, B.A., Young, S., 2018a. Role of the foredune in controlling barrier island response to sea level rise. In: Moore, L.J., Murray, A.S. (Eds.), *Barrier Dynamics and Response to Climate Change*. Springer.
- Houser, C., Wernette, P., Weymer, B.A., 2018b. Scale-dependent behavior of the foredune: Implications for barrier island response to storms and sea-level rise. *Geomorphology* 303, 362–374.
- Hugenholtz, C.H., Wolfe, S.A., Walker, I.J., Moorman, B.J., 2009. Spatial and temporal patterns of Aeolian sediment transport on an inland parabolic dune, Bigstick Sand Hills, Saskatchewan, Canada. *Geomorphology* 105 (1–2), 158–170.
- Johannessen, O.M., Bengtsson, L., Miles, M.W., Kuzmina, S.I., Semenov, V.A., Alekseev, G.V., Nagurnyi, A.P., Zakharov, V.F., Bobylev, L.P., Pettersson, L.H., Hasselmann, K., Cattle, H. P., 2016. Arctic climate change: observed and modelled temperature and sea-ice variability. *Tellus Ser. A Dyn. Meteorol. Oceanogr.* 56 (4), 328–341.
- Kwok, R., Rothrock, D.A., 2009. Decline in Arctic sea ice thickness from submarine and ICESat records: 1958–2008. *Geophys. Res. Lett.* 36 (15) (n/a–n/a).
- Lentz, E.E., Hapke, C.J., Stockdon, H.F., Hehre, R.E., 2013. Improving understanding of near-term barrier island evolution through multi-decadal assessment of morphologic change. *Mar. Geol.* 337, 125–139.
- Lindsay, R.W., Zhang, J., 2005. The thinning of Arctic sea ice, 1988–2003: have we passed a tipping point. *J. Clim.* 18, 4879–4894.
- Mathew, S., Davidson-Arnott, R.G.D., Ollerhead, J., 2010. Evolution of a beach–dune system following a catastrophic storm overwash event: Greenwich Dunes, Prince Edward Island, 1936–2005. *Can. J. Earth Sci.* 47 (3), 273–290.
- Roberts, T.M., Wang, P., 2012. Four-year performance and associated controlling factors of several beach nourishment projects along three adjacent barrier islands, west-central Florida, USA. *Coast. Eng.* 70, 21–39.
- Scott, T., Masselink, G., O'Hare, T., Sauter, A., Poate, T., Russell, P., Davidson, M., Conley, D., 2016. The extreme 2013–2014 winter storms: beach recovery along the southwest coast of England. *Mar. Geol.* 382, 224–241.
- Sherman, D.J., Hales, B.U., Potts, M.K., Ellis, J.T., Liu, H., Houser, C., 2013. Impacts of Hurricane Ike on the beaches of the Bolivar Peninsula, TX, USA. *Geomorphology* 199, 62–81.
- Splinter, K.D., Kearney, E.T., Turner, I.L., 2018. Drivers of alongshore variable dune erosion during a storm event: observations and modelling. *Coast. Eng.* 131, 31–41.
- Stammerjohn, S., Massom, R., Rind, D., Martinson, D., 2012. Regions of rapid sea ice change: an inter-hemispheric seasonal comparison. *Geophys. Res. Lett.* 39 (6) (n/a–n/a).
- Su, J., Bork, E., 2006. Influence of vegetation, slope, and LiDAR sampling angle on DEM accuracy. *Photogramm. Eng. Remote. Sens.* 72 (11), 1265–1274.
- Sutherland, J., 2012. Error analysis of Ordnance Survey map tidelines, UK. *Proc. ICE Marit. Eng.* 165 (4), 189–197.
- Walker, I.J., Nickling, W.G., 2003. Simulation and measurement of surface shear stress over isolated and closely spaced transverse dunes in a wind tunnel. *Earth Surf. Process. Landf.* 28 (10), 1111–1124.
- Walker, I.J., Eamer, J.B.R., Darke, I.B., 2013. Assessing significant geomorphic changes and effectiveness of dynamic restoration in a coastal dune ecosystem. *Geomorphology* 199, 192–204.
- Weymer, B.A., Houser, C., Giardino, J.R., 2015. Poststorm evolution of beach–dune morphology: Padre Island National Seashore, Texas. *J. Coast. Res.* 31 (3), 634–644.
- Weymer, B.A., Everett, M.E., de Smet, T.S., Houser, C., 2015b. Review of electromagnetic induction for mapping barrier island framework geology. *Sediment. Geol.* 321, 11–24.
- Weymer, B., Everett, M.E., Houser, C., Wernette, P., Barrineau, P., 2016. Differentiating tidal and groundwater dynamics from barrier island framework geology: testing the utility of portable multifrequency electromagnetic induction profilers. *Geophysics* 81 (5), E347–E361.
- Wheaton, J.M., Brasington, J., Darby, S.E., Sear, D.A., 2010. Accounting for uncertainty in DEMs from repeat topographic surveys: improved sediment budgets. *Earth Surf. Process. Landf.* 35, 136–156.